Robot Scheduling Control System (RSC)

Local Dispatch

Task Management

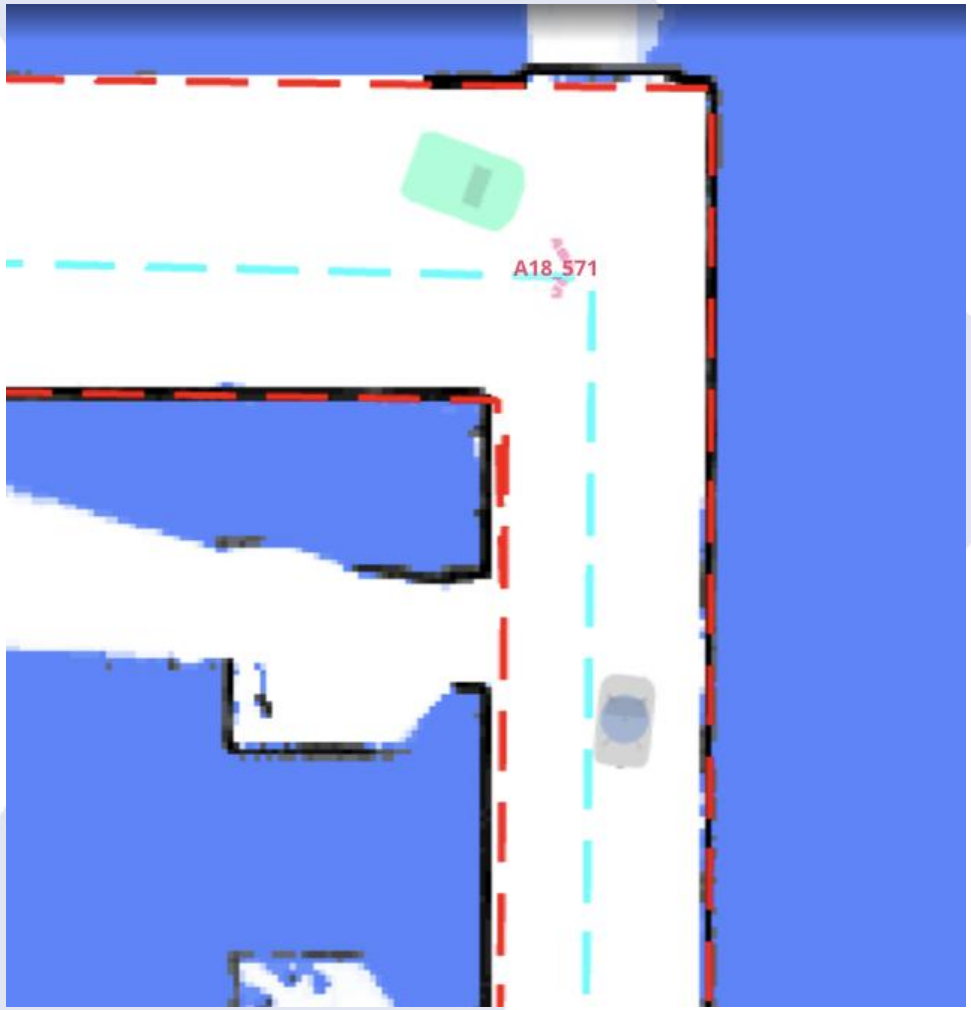
Route Planning

Status Monitoring

System Integration



Robot Scheduling Control System



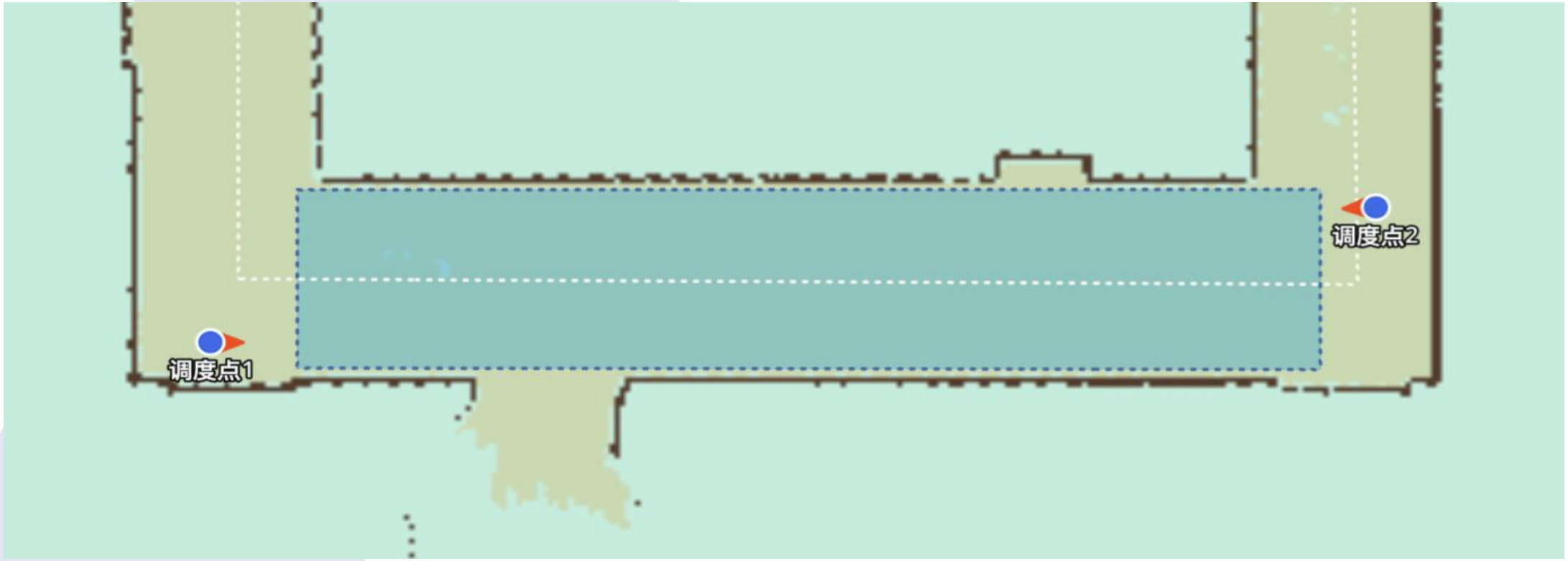
V2X, short for Vehicle to Everything, is a wireless communication technology designed for vehicles. Its purpose is to enable information exchange between a vehicle and any surrounding entities that might affect it, including but not limited to other vehicles, pedestrians, infrastructure (such as traffic lights), and external networks.

An AGV (Automated Guided Vehicle) equipped with a V2X module can use advanced wireless communication technology to exchange critical information with other vehicles in real time, such as location, speed, and driving direction. Based on this shared information, the AGV can perform precise calculations and judgments, effectively avoiding collision risks and achieving synchronized path planning. The application of this technology significantly enhances driving safety and provides a more reliable and efficient solution for automated logistics transportation.

Robot Scheduling Control System

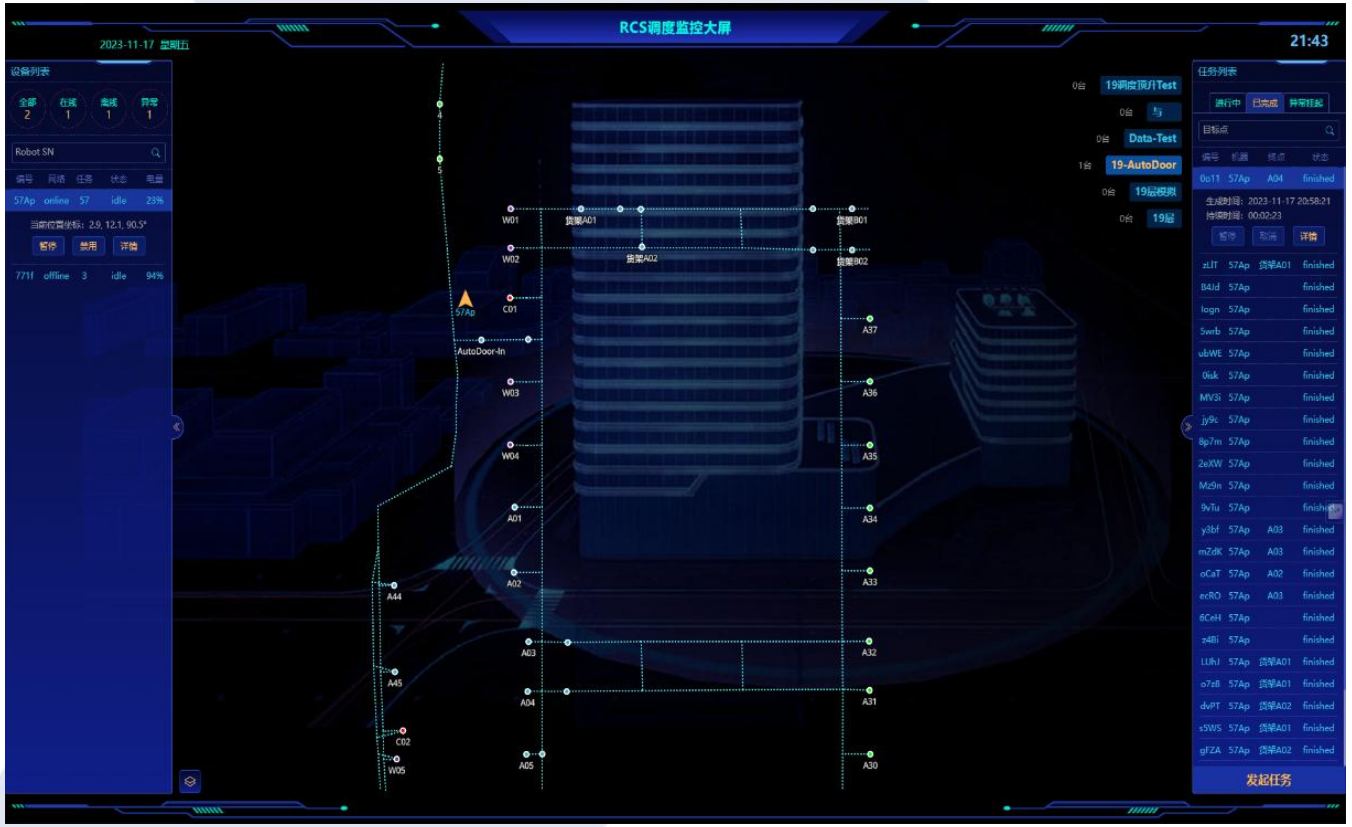
With the support of the V2X communication module, more detailed handling of certain specialized operational scenarios can be achieved. For example, in narrow passage scenarios where the path is too tight for vehicles to pass each other, the issue can be perfectly resolved by adding narrow passage areas and corresponding dispatch points in the map editor.

After adding a narrow passage area, only one vehicle is allowed to enter the area at a time. If other vehicles need to pass through, they will wait in line at the dispatch point until they can proceed.



Robot Scheduling Control System

RCS Robot Scheduling Control System is an important component in the field of mobile robotics, which is responsible for the robot's task assignment, path planning, status monitoring, and collaborative work with other systems. Through the use of AI technologies such as machine learning and multi-intelligence gaming, the RCS system ensures that robots perform tasks efficiently and accurately in complex environments.



Multi-model mixed scheduling: support multiple type robot scheduling, backpack, lift, submerged, forklift and other multi-model mixed use; load-balanced cluster deployment, support multiple RCS service lateral expansion;

Path planning algorithms: support for shortest path planning, traffic dynamic management, avoidance, re-planning, and support for very large map planning;

Status monitoring: Robot status monitoring, real-time monitoring of the robot's online status, load status, work status, position coordinates, battery power and other information.

Abnormal handling: timely detection and handling of abnormalities in the operation of the robot to ensure stable operation of the system.

Multiple Peripheral Docking: Support business scenarios containing peripherals such as caller, automatic doors, gates, lifts, indicator, sound, etc;

Multiple interfacing methods: Provide standard external interfaces for upper-level systems (MES/WMS, etc.), multiple communication protocols coexist, and meet a variety of in-plant logistics handling needs.

RCS Robot Scheduling Control System

It is used to monitor the status of AGV equipment and task execution, divided into functional modules such as operation control, control intervention, alerts, and task monitoring. The monitoring client allows users and maintenance personnel to promptly detect anomalies in AGV operations and tasks, enabling proactive intervention to ensure the smooth operation of logistics activities.

Video recording and playback

Supports interface recording during operation and playback of recorded sessions.

Real-time Operation Display

Real-time updates of AMR position, including device battery levels, network quality, and task execution monitoring, with support for route display.

Operation Alarm Monitoring

Includes obstacle detection, offline status, task verification, and other alerts.

Convenient Remote Maintenance

Supports remote layout adjustments and capacity expansion, enabling quick system deployment and operation.



① **Common Functions: Dynamic map path display, roadblock editing,** and other functionalities.

② **Cyclic Charging Function**: Based on a configurable **battery threshold management strategy**, it monitors the AGV's battery status in real time. When the AGV's battery falls below the preset threshold, the system automatically plans a route to the charging station and assigns a charging task. Once charging is complete, the AGV will **automatically return to its working state** to continue executing unfinished tasks or to receive new tasks.

③ **Statistical Functions**: Includes **statistics** on average AGV battery levels, storage position counts, equipment status, and tasks in the platform and equipment, such as idle, charging, and active tasks.

④ **Warning Functions**: Displays current system warnings and provides guidance on how to resolve corresponding errors, helping users to clear the warnings and ensure normal system operation.

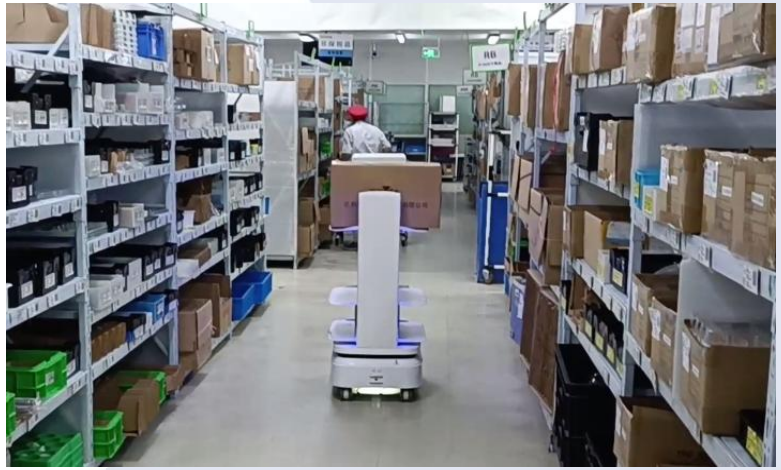
⑤ **Grouping and Zoning**: Based on the spatial layout of the warehouse or work area, zones are divided reasonably to ensure that robots can **operate efficiently** within their designated areas. This avoids unnecessary cross-zone movements and ensures that the workload among robots in different groups or zones is **relatively balanced**, preventing situations where robots in some areas are overworked while others are idle.

RCS Robot Scheduling Control System - Common Handling Scenarios

- **Task Suspension:** When the target area or target point of a task has no available storage locations or no suitable racks, making it impossible to carry out the handling task, the system can suspend the task. The system will continuously monitor the situation, and once the conditions for handling are met, the task will be executed automatically without the need for reactivation.
- **Traffic Management:** Coordinates the traffic between multiple robots to ensure smooth flow and avoid conflicts and waiting times between robots.
- **Outbound Order:** A task triggered by a workstation is treated as a task group. The RCS scheduling system executes each task in the group strictly according to the sequence defined by the higher-level system. This means that the order in which AGVs transport target racks to the workstation follows the sequence specified by the higher-level system.
- **Queue Number Setting:** There is no upper limit on the number of tasks created by the higher-level system. Users can pre-set an unlimited number of handling tasks without manual task editing, thus freeing up human resources from waiting.
- **Different Priorities for Task Types:** If the project involves multiple types of tasks, priority settings can be used to determine which type of task should be executed first when AGV resources are insufficient.

RCS Robot Scheduling Control System - Common Handling Scenarios

E-commerce Sector: In e-commerce warehouses, the RCS system can manage multiple robots for operations such as inventory management, inbound and outbound logistics, and stocktaking. This enhances warehouse management efficiency and accuracy.



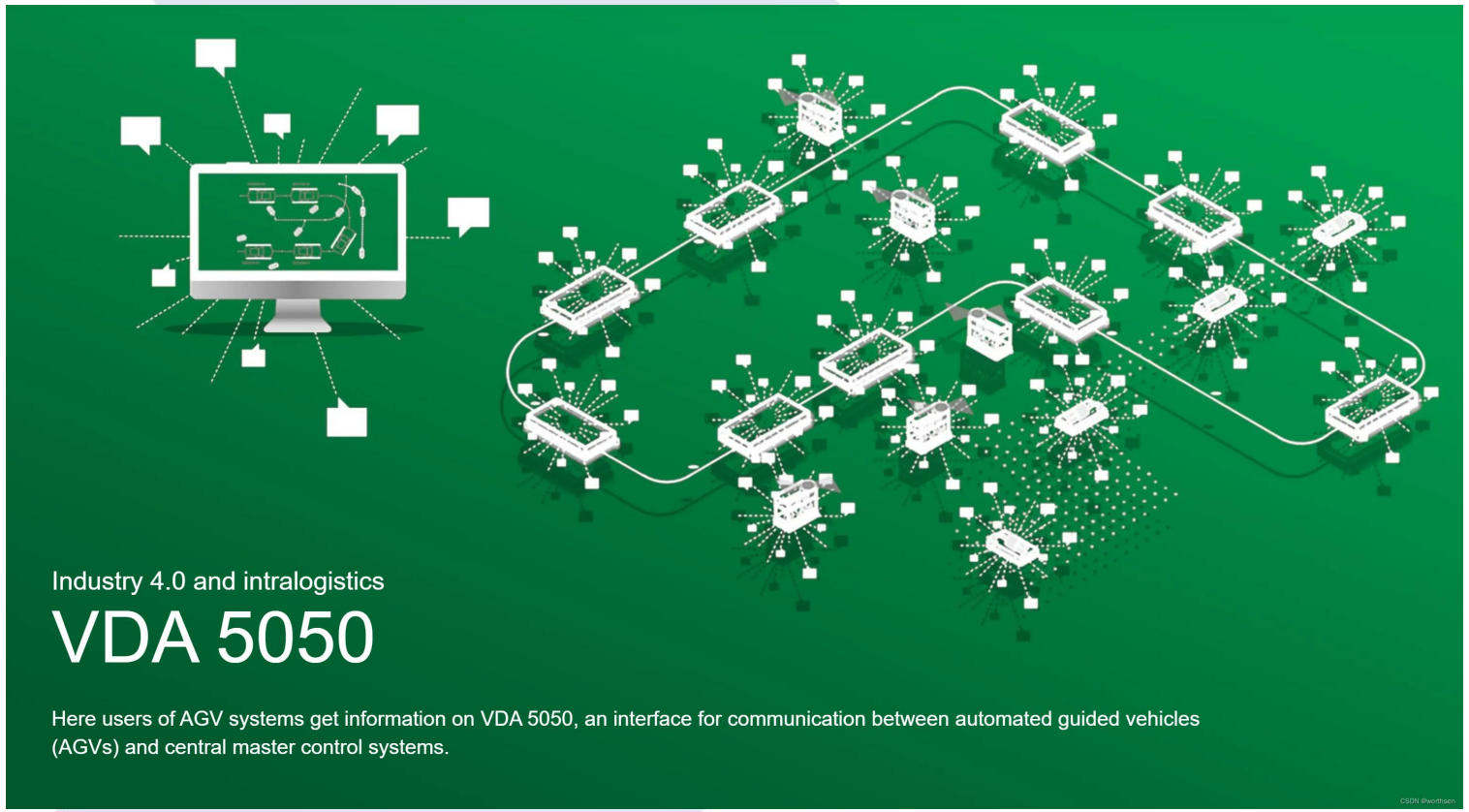
3C Electronics: Integrates with factory operations to improve the efficiency of conveyor belt material handling and reduce error rates due to human intervention.



Logistics Sector: In large logistics warehouses, the RCS system can manage multiple AGVs to achieve automated material handling and sorting, thereby enhancing logistics efficiency.

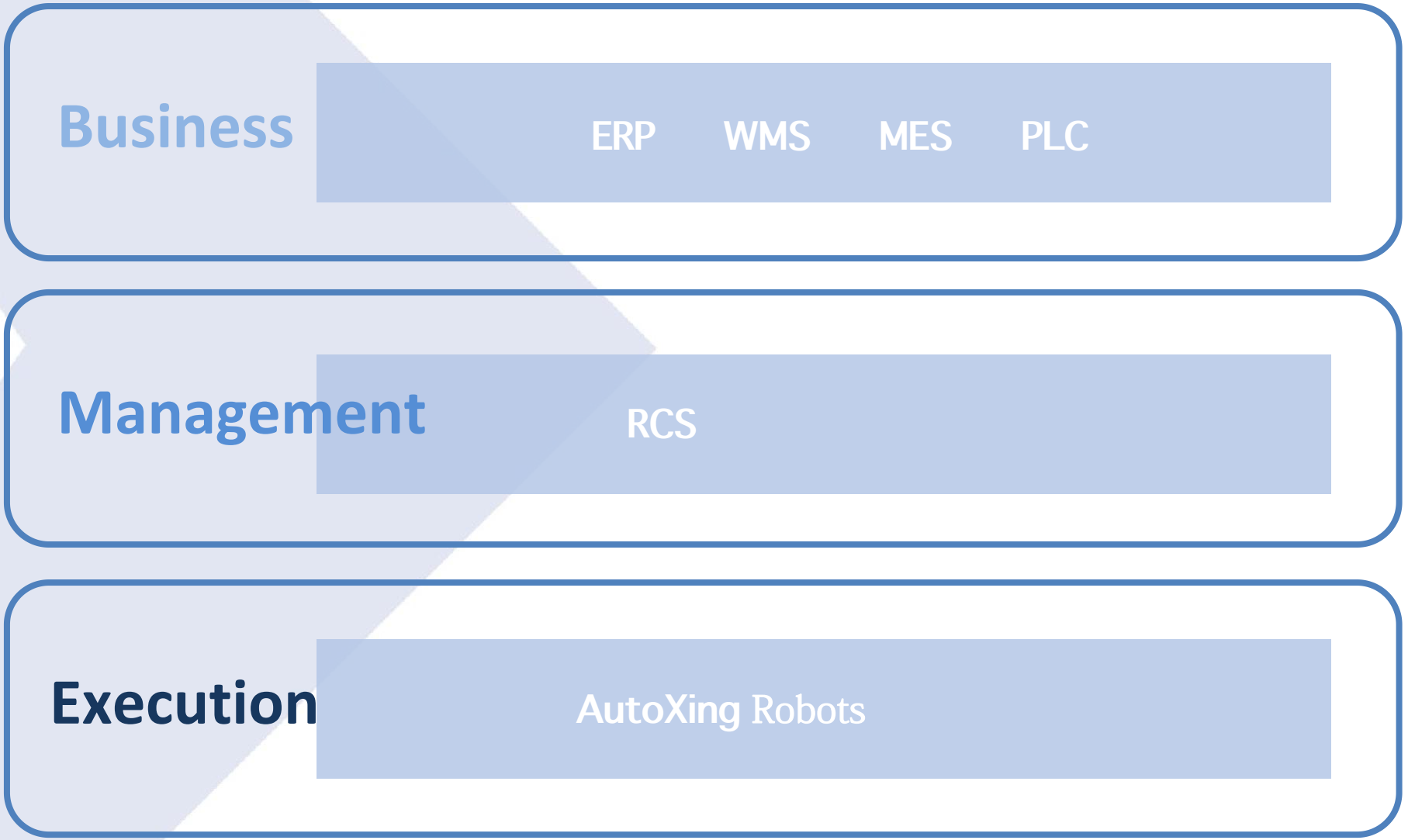


RCS Robot Scheduling Control System - vds5050



The VDA5050 protocol is a standard communication interface for mobile robots (AGVs/AMRs) developed through the collaboration of the German Association of the Automotive Industry (VDA) and the German Engineering Federation (VDMA). It uses the MQTT standard messaging protocol to facilitate communication. The VDA5050 protocol aims to standardize communication between mobile robots and scheduling systems (main control), enabling seamless integration of different brands and types of mobile robots within the same scheduling system. This facilitates mixed scheduling and collaborative operations, accelerating the mobile robot industry's move towards standardization and efficiency.

RCS Robot Scheduling Control System - System Integration



RCS Robot Scheduling Control System - System Integration

System Flexibility to Meet Personalized Needs

RCS can seamlessly integrate with various enterprise systems such as WMS, MES, and ERP. For different business scenarios, RCS offers standard interfaces for both warehousing and handling integration modes, demonstrating high compatibility and scalability.

It also supports customized requirements and server expansion deployment.



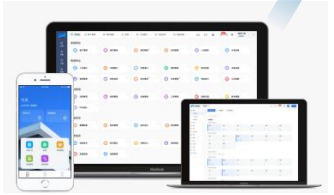
Operation Monitor Screen



Cloud Scheduling System

Smart Platform

Full System Connectivity,
Complete Data Integration



SaaS



Integration with Factory Business Systems, Production ERP, and Logistics WMS

Thank You